

LAPORAN PRAKTIKUM JOBSHEET 1

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Kelas: TI-2C

Absen: 17

1. Prosedural Programming

Kode:

```
package Latihan;

public class Prosedural{
    public static void main(String[] args) {
        String itemName1 = "Coffee";
        float price = 120000.0f;
        int quantity1 = 3;

        String itemName2 = "Sugar";
        float price2 = 30000.0f;
        int quantity2 = 2;

        String itemName3 = "Matcha Latte";
        float price3 = 15000.0f;
        int quantity3 = 4;

        System.out.println(itemName1 + " : " + price + " x " +
quantity1 + " = " + calculateTotal(price, quantity1));
        System.out.println(itemName2 + " : " + price2 + " x " +
quantity2 + " = " + calculateTotal(price2, quantity2));
        System.out.println(itemName3 + " : " + price3 + " x " +
quantity3 + " = " + calculateTotal(price3, quantity3));
    }

    public static float calculateTotal(float price, int quantity)
    {
        return price * quantity;
    }
}
```

Hasil:

```
2 (bin) Latihan:Prosedural
Coffee : 120000.0 x 3 = 360000.0
Sugar : 30000.0 x 2 = 60000.0
Matcha Latte : 15000.0 x 4 = 60000.0
```

2. Object Oriented Programming

Kode:

```
package Latihan;

public class OOP {
    public static void main(String[] args) {
        Product product1 = new Product("Coffee", 120000.0f,
"Starbucks");
        Product product2 = new Product("Sugar", 30000.0f,
"Tiger");

        System.out.println(product1.getItemName() + " : " +
product1.total(3) + " (Brand: " + product1.getBrand() + ")");
        System.out.println(product2.getItemName() + " : " +
product2.total(2) + " (Brand: " + product2.getBrand() + ")");
    }
}

class Product{
    private final String itemName;
    private final float price;
    private final String brand;

    public Product(String itemName, float price, String brand) {
        this.itemName = itemName;
        this.price = price;
        this.brand = brand;
    }

    public float total(int quantity) {
        return price * quantity;
    }

    public String getItemName() {
        return itemName;
    }

    public String getBrand() {
        return brand;
    }
}
```

Hasil:

```
2\bin' 'Latihan.OOP'  
Coffee : 360000.0 (Brand: Starbucks)  
Sugar : 60000.0 (Brand: Tiger)  
D:\Users\rezia\Documents\Semester 2\15
```

3. Percobaan dan modifikasinya

Kode di Bike.java

```
package Percobaan;  
  
public class Bike {  
    private String brand;  
    private int speed;  
    private int gear=1;  
    public int distance;  
    //gear 1: max5km/h, gear 2: max10km/h, gear 3: max15km/h  
    private final int[] gearSpeedLimits = {5, 10, 25, 30, 40,  
60};  
  
    public void setBrand(String brandName) {  
        brand = brandName;  
    }  
  
    public void gearChanges(int gearValue) {  
        if (gearValue >= 1 && gearValue <= 6) {  
            gear = gearValue;  
        } else {  
            System.out.println("Invalid gear. Please select a  
gear between 1 and 6.");  
        }  
    }  
  
    public int speedAcceleration(int increment) {  
        speed += increment;  
        if (speed > gearSpeedLimits[gear - 1]) {  
            speed = gearSpeedLimits[gear - 1];  
        }  
        return speed;  
    }  
  
    public int speedDeceleration(int decrement) {  
        speed -= decrement;  
        if (speed < 0) {  
            speed = 0;  
        }  
    }  
}
```

```

        return speed;
    }

    public int jarak(int time) {
        distance = speed * time;
        return distance;
    }

    public void printInfo() {
        System.out.println("Brand: " + brand);
        System.out.println("Current Speed: " + speed + " km/h");
        System.out.println("Current Gear: " + gear);
        System.out.println("Distance Traveled: " + distance + "
km");
    }
}

```

kode di BikeDemo.java:

```

package Percobaan;

public class bikeDemo {
    public static void main(String[] args) {
        Bike bike1 = new Bike();
        bike1.setBrand("Polygon");
        bike1.gearChanges(2);
        bike1.speedAcceleration(8);
        bike1.speedDeceleration(3);
        bike1.jarak(2); // Calculate distance traveled in 2 hours
        bike1.printInfo();

        System.out.println();

        Bike bike2 = new Bike();
        bike2.setBrand("United");
        bike2.gearChanges(3);
        bike2.speedAcceleration(12);
        bike2.jarak(1); // Calculate distance traveled in 1 hour
        bike2.speedDeceleration(2);
        bike2.printInfo();
    }
}

```

```

        System.out.println();

        RoadBike roadBike1 = new RoadBike();
        roadBike1.setBrand("Specialized");
        roadBike1.setTireWidth("25");
        roadBike1.gearChanges(4);
        roadBike1.speedDeceleration(5);
        roadBike1.speedAcceleration(20);
        roadBike1.jarak(3); // Calculate distance traveled in 1
hour
        roadBike1.printInfo();
    }
}

```

Kode di RoadBike.java:

```

package Percobaan;

public class RoadBike extends Bike {
    private String tireWidth;

    public void setTireWidth(String width) {
        tireWidth = width;
    }

    @Override
    public void printInfo() {
        super.printInfo();
        System.out.println("Tire Width: " + tireWidth + " mm");
        System.out.println("Bike Type: Road Bike");
    }
}

```

Hasil:

```

2\bin' 'Percobaan.bikeDemo'
Brand: Polygon
Current Speed: 5 km/h
Current Gear: 2
Distance Traveled: 10 km

Brand: United
Current Speed: 10 km/h
Current Gear: 3
Distance Traveled: 12 km

Brand: Specialized
Current Speed: 20 km/h
Current Gear: 4
Distance Traveled: 60 km
Tire Width: 25 mm
Bike Type: Road Bike
D5-G:\Users\rezia\Documents\

```

5. Pertanyaan

1. Jelaskan perbedaan antara object dengan class!
2. Jelaskan alasan gear dan brand dapat menjadi atribut dari object Bike!
3. Sebutkan salah satu kelebihan utama dari pemrograman berorientasi objek dibandingkan dengan pemrograman prosedural!
4. Apakah diperbolehkan melakukan pendefinisian dua buah atribut dalam satu baris kode seperti "public String nama, alamat;"?
5. Pada class RoadBike, jelaskan alasan atribut brand, speed, dan gear tidak lagi ditulis di dalam class tersebut!

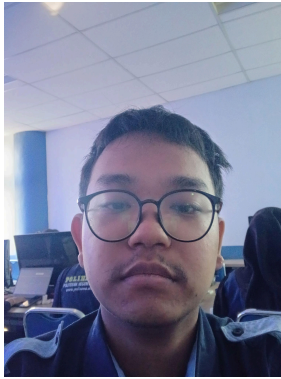
Jawaban:

1. Class adalah template atau blueprint dari object sedangkan object adalah hasil blueprint itu.
2. Karne gear dan brand kita deklarasikan sebagai attribute object Bike.
3. Mengurangi repetisi, dan membuat kode lebih mudah di track dan dibaca.
4. Boleh
5. Karena objetc RoadBike sudah mewarisi attribut tersebut.

6. Tugas Praktikum

Foto object:





Object di atas ada sepatu, laptop, botol, dan manusia.

1. Object sepatu

Attribut:

```
private String merk;  
    private String warna;  
    private String ukuran;  
    private int harga;  
    public int jumlah;
```

Method:

```
tampilkanInfo()  
getUkuranSepatu()  
getHargaSepatu()  
getMerk()  
getWarna()  
    getJumlah()  
ubahWarna(String warnaBaru)  
ubahUkuran(String ukuranBaru)
```

Full code:

```
package Tugas;  
  
public class Sepatu{  
    public static void main(String[] args) {  
        Barang sepatu1 = new Barang("Nike", "Hitam", "42",  
800000);  
  
        sepatu1.tampilkanInfo();  
        System.out.println();  
  
        System.out.println("Jumlah: " + sepatu1.getJumlah());  
        System.out.println("Merk: " + sepatu1.getMerk());  
    }  
}
```

```

        System.out.println("Warna: " + sepatu1.getWarna());
        System.out.println("Ukuran: " +
sepatu1.getUkuranSepatu());

        System.out.println(sepatu1.ubahWarna("Putih"));
        System.out.println(sepatu1.ubahUkuran("43"));

    }
}

class Barang{
    private String merk;
    private String warna;
    private String ukuran;
    private int harga;
    public int jumlah;

    public Barang(String merk, String warna, String ukuran, int
harga) {
        this.merk = merk;
        this.warna = warna;
        this.ukuran = ukuran;
        this.harga = harga;
        jumlah++;
    }

    public void tampilkanInfo() {
        System.out.println("Merk: " + merk);
        System.out.println("Warna: " + warna);
        System.out.println("Ukuran: " + ukuran);
        System.out.println("Harga: Rp" + harga);
    }

    public String getUkuranSepatu(){
        return ukuran;
    }

    public int getHargaSepatu(){
        return harga;
    }

    public String getMerk() {
        return merk;
    }
}

```

```

    }

    public String getWarna() {
        return warna;
    }

    public int getJumlah() {
        return jumlah;
    }

    public String ubahWarna(String warnaBaru) {
        this.warna = warnaBaru;
        return warna;
    }

    public String ubahUkuran(String ukuranBaru) {
        this.ukuran = ukuranBaru;
        return ukuran;
    }
}

```

Hasil:

```

Merk: Nike
Warna: Hitam
Ukuran: 42
Harga: Rp800000

Jumlah: 1
Merk: Nike
Warna: Hitam
Ukuran: 42
Putih
43

```

2. Object Laptop:

Attribut:

```

private String merk;
private String processor;
private int ram;
private int harga;
private boolean isAvailable;
private int tambahRam;

```

Method:

```
tambahRam(int tambahan)
tampilkanInfo()
isAvailable()
```

Full code:

```
package Tugas;

public class Laptop {
    public static void main(String[] args) {
        Product laptop1 = new Product("Asus", "Intel Core i7",
16, 15000000, true, 8);
        Product laptop2 = new Product("Acer", "Intel Core i5", 8,
10000000, false, 4);

        laptop1.tampilkanInfo();
        System.out.println();

        laptop2.tampilkanInfo();
    }
}

class Product {
    private String merk;
    private String processor;
    private int ram;
    private int harga;
    private boolean isAvailable;
    private int tambahRam;

    public Product(String merk, String processor, int ram, int
harga, boolean isAvailable, int tambahanRam) {
        this.merk = merk;
        this.processor = processor;
        this.ram = ram;
        this.harga = harga;
        this.isAvailable = isAvailable;
        this.tambahRam = tambahanRam;
    }

    public int tambahRam(int tambahan) {
        this.ram += tambahRam;
        return ram;
    }
}
```

```

    }

    public void tampilkanInfo() {
        System.out.println("Merk: " + merk);
        System.out.println("Processor: " + processor);
        System.out.println("RAM: " + ram + " GB");
        System.out.println("Harga: Rp" + harga);
    }

    public boolean isAvailable() {
        return isAvailable;
    }
}

```

Hasil:

```

Merk: Asus
Processor: Intel Core i7
RAM: 16 GB
Harga: Rp15000000

Merk: Acer
Processor: Intel Core i5
RAM: 8 GB
Harga: Rp10000000

```

3. Object botol

Attribut:

```

private int width;
    private int kapasitas;
    private int harga;
    private String kegunaan;

```

Method:

```

tampilkanInfo()
getWidth()
getHarga
getKapasitas
getKegunaan
setWidth(int width)
setKapasitas(int kapasitas)
setHarga(int harga)
setKegunaan(String kegunaan)

```

Full code:

```
package Tugas;

public class botol {
    public static void main(String[] args) {
        Botol botol1 = new Botol(10, 600, 5000, "diminum");
        botol1.tampilkanInfo();
    }
}

class Botol {
    private int width;
    private int kapasitas;
    private int harga;
    private String kegunaan;

    public Botol(int width, int kapasitas, int harga, String
kegunaan) {
        this.width = width;
        this.kapasitas = kapasitas;
        this.harga = harga;
        this.kegunaan = kegunaan;
    }

    public void tampilkanInfo() {
        System.out.println("Width: " + width);
        System.out.println("Kapasitas: " + kapasitas + " ml");
        System.out.println("Harga: Rp" + harga);
        System.out.println("Kegunaan: " + kegunaan);
    }

    public int getWidth() {
        return width;
    }

    public int getKapasitas() {
        return kapasitas;
    }

    public int getHarga() {
        return harga;
    }

    public String getKegunaan() {
        return kegunaan;
    }
}
```

```

    public void setWidth(int width) {
        this.width = width;
    }
    public void setKapasitas(int kapasitas) {
        this.kapasitas = kapasitas;
    }
    public void setHarga(int harga) {
        this.harga = harga;
    }
    public void setKegunaan(String kegunaan) {
        this.kegunaan = kegunaan;
    }
}

```

Object: Aqua extend botol

attribut:

```

private String warna;
    private String merk;
    public int tinggiBotol;
    public String rasa;

```

method:

```

tampilkanInfo
gantiRasa
gantiWarna

```

Full code:

```

package Tugas;

public class aqua extends Botol {
    private String warna;
    private String merk;
    public int tinggiBotol;
    public String rasa;

    public aqua(String warna, String merk, int tinggiBotol,
String kegunaan, int width, int kapasitas, int harga, String
rasa) {
        super(width, kapasitas, harga, kegunaan);
        this.warna = warna;
        this.merk = merk;
    }
}

```

```

        this.tinggiBotol = tinggiBotol;
        this.rasa = rasa;
    }

    public void tampilkanInfo() {
        super.tampilkanInfo();
        System.out.println("Warna: " + warna);
        System.out.println("Merk: " + merk);
        System.out.println("Tinggi Botol: " + tinggiBotol + "
cm");
        System.out.println("Rasa: " + rasa);
        System.out.println("Botol Aqua ini memiliki warna " +
warna + ", merk " + merk + ", tinggi botol " + tinggiBotol + "
cm, dan rasa " + rasa + ".");
    }

    public String gantiRasa(String rasaBaru) {
        this.rasa = rasaBaru;
        return "Rasa baru: " + rasa;
    }

    public String gantiWarna(String warnaBaru) {
        this.warna = warnaBaru;
        return "Warna baru: " + warna;
    }
}

```

Full code setelah digabung Aqua:

```

package Tugas;

public class botol {
    public static void main(String[] args) {
        Botol botol1 = new Botol(10, 600, 5000, "diminum");
        botol1.tampilkanInfo();

        System.out.println();
        botol1.setWidth(12);
        botol1.setKapasitas(750);
        botol1.setHarga(6000);
        botol1.setKegunaan("minum");
        botol1.tampilkanInfo();
    }
}

```

```

        System.out.println();
        aqua aqual = new aqua("Biru", "Aqua", 20, "diminum", 10,
600, 5000, "Original");
        aqual.tampilkanInfo();

        System.out.println();
        aqual.gantiRasa("Jeruk");
        aqual.gantiWarna("Hijau");
        System.out.println("Setelah mengganti rasa dan warna:");
        aqual.tampilkanInfo();

    }
}

class Botol {
    private int width;
    private int kapasitas;
    private int harga;
    private String kegunaan;

    public Botol(int width, int kapasitas, int harga, String
kegunaan) {
        this.width = width;
        this.kapasitas = kapasitas;
        this.harga = harga;
        this.kegunaan = kegunaan;
    }

    public void tampilkanInfo() {
        System.out.println("Width: " + width);
        System.out.println("Kapasitas: " + kapasitas + " ml");
        System.out.println("Harga: Rp" + harga);
        System.out.println("Kegunaan: " + kegunaan);
    }

    public int getWidth() {
        return width;
    }

    public int getKapasitas() {
        return kapasitas;
    }

    public int getHarga() {

```

```

        return harga;
    }

    public String getKegunaan() {
        return kegunaan;
    }

    public void setWidth(int width) {
        this.width = width;
    }

    public void setKapasitas(int kapasitas) {
        this.kapasitas = kapasitas;
    }

    public void setHarga(int harga) {
        this.harga = harga;
    }

    public void setKegunaan(String kegunaan) {
        this.kegunaan = kegunaan;
    }
}

```

hasil run botol:

```

C:\Users\roziq\Documents> java Botol
Width: 10
Kapasitas: 600 ml
Harga: Rp5000
Kegunaan: diminum

```

```

C:\Users\roziq\Documents> java Botol 12 750 6000 minum
Width: 12
Kapasitas: 750 ml
Harga: Rp6000
Kegunaan: minum

```

hasil run aqua:

```

C:\Users\roziq\Documents> java Aqua
Width: 10
Kapasitas: 600 ml
Harga: Rp5000
Kegunaan: diminum
Warna: Biru
Merk: Aqua
Tinggi Botol: 20 cm
Rasa: Original
Botol Aqua ini memiliki warna Biru, merk Aqua, tinggi botol 20 cm, dan rasa Original.

```

```
Setelah mengganti rasa dan warna:  
Width: 10  
Kapasitas: 600 ml  
Harga: Rp5000  
Kegunaan: diminum  
Warna: Hijau  
Merk: Aqua  
Tinggi Botol: 20 cm  
Rasa: Jeruk  
Botol Aqua ini memiliki warna Hijau, merk Aqua, tinggi botol 20 cm, dan rasa Jeruk.  
PS C:\Users\cozig\Documents\Semester3\PRO\Praktikum\Jobsheet1> █
```

4. Object manusia:

Attribut:

```
private String nama;  
    private int umur;  
    private String jenisKelamin;  
    private String pekerjaan;
```

Method:

```
tampilkanInfo  
tambahUmur  
gantiPekerjaan
```

Full Code:

```
package Tugas;  
  
public class Manusia {  
  
    public static void main(String[] args) {  
        Character character1 = new Character("John", 25,  
"Laki-laki", "Programmer");  
        character1.tampilkanInfo();  
  
        System.out.println();  
        character1.gantiPekerjaan("Designer");  
        character1.tambahUmur(1);  
        character1.tampilkanInfo();  
    }  
}  
  
class Character {  
    private String nama;  
    private int umur;  
    private String jenisKelamin;
```

```

        private String pekerjaan;

        public Character(String nama, int umur, String jenisKelamin,
String pekerjaan) {
            this.nama = nama;
            this.umur = umur;
            this.jenisKelamin = jenisKelamin;
            this.pekerjaan = pekerjaan;
        }

        public void tampilkanInfo() {
            System.out.println("Nama: " + nama);
            System.out.println("Umur: " + umur);
            System.out.println("Jenis Kelamin: " + jenisKelamin);
            System.out.println("Pekerjaan: " + pekerjaan);
        }

        public String gantiPekerjaan(String pekerjaanBaru) {
            this.pekerjaan = pekerjaanBaru;
            return "Pekerjaan baru: " + pekerjaan;
        }

        public int tambahUmur(int tambahan) {
            this.umur += tambahan;
            return umur;
        }
    }
}

```

Hasil:

```

PS C:\Users\rozig\Documents\Seri
Nama: John
Umur: 25
Jenis Kelamin: Laki-laki
Pekerjaan: Programmer

Nama: John
Umur: 26
Jenis Kelamin: Laki-laki
Pekerjaan: Designer

```

Object Michiko(teman saya):

Attribute:

```
private String hobi;  
private String ras;  
private String status;  
private String agama;
```

Method:

```
tampilkanInfo  
gantiRas  
gantiStatus
```

Full code:

```
package Tugas;  
  
public class Michiko extends Character {  
    private String hobi;  
    private String ras;  
    private String status;  
    private String agama;  
  
    public Michiko(String nama, int umur, String jenisKelamin,  
String pekerjaan, String hobi, String ras, String status, String  
agama) {  
        super(nama, umur, jenisKelamin, pekerjaan);  
        this.hobi = hobi;  
        this.ras = ras;  
        this.status = status;  
        this.agama = agama;  
    }  
  
    public void tampilkanInfo() {  
        super.tampilkanInfo();  
        System.out.println("Hobi: " + hobi);  
        System.out.println("Ras: " + ras);  
        System.out.println("Status: " + status);  
        System.out.println("Agama: " + agama);  
    }  
  
    public String gantiHobi(String hobiBaru) {  
        this.hobi = hobiBaru;  
    }  
}
```

```

        return "Hobi baru: " + hobi;
    }

    public String gantiRas(String rasBaru) {
        this.ras = rasBaru;
        return "Ras baru: " + ras;
    }

    public String gantiStatus(String statusBaru) {
        this.status = statusBaru;
        return "Status baru: " + status;
    }
}

```

kode di manusia.java:

```

package Tugas;

public class Manusia {

    public static void main(String[] args) {
        Character character1 = new Character("John", 25,
        "Laki-laki", "Programmer");
        character1.tampilkanInfo();

        System.out.println();
        character1.gantiPekerjaan("Designer");
        character1.tambahUmur(1);
        character1.tampilkanInfo();

        System.out.println();
        Michiko michiko = new Michiko("Michiko", 20, "Perempuan",
        "Student", "Reading", "Japanese", "Single", "Buddha");
        michiko.tampilkanInfo();

        System.out.println();
        michiko.gantiHobi("Traveling");
        michiko.gantiRas("Korean");
        michiko.gantiStatus("Married");
        michiko.tampilkanInfo();
    }
}

```

```

class Character {
    private String nama;
    private int umur;
    private String jenisKelamin;
    private String pekerjaan;

    public Character(String nama, int umur, String jenisKelamin,
String pekerjaan) {
        this.nama = nama;
        this.umur = umur;
        this.jenisKelamin = jenisKelamin;
        this.pekerjaan = pekerjaan;
    }

    public void tampilkanInfo() {
        System.out.println("Nama: " + nama);
        System.out.println("Umur: " + umur);
        System.out.println("Jenis Kelamin: " + jenisKelamin);
        System.out.println("Pekerjaan: " + pekerjaan);
    }

    public String gantiPekerjaan(String pekerjaanBaru) {
        this.pekerjaan = pekerjaanBaru;
        return "Pekerjaan baru: " + pekerjaan;
    }

    public int tambahUmur(int tambahan) {
        this.umur += tambahan;
        return umur;
    }
}

```

Hasil:

Nama: Michiko
Umur: 20
Jenis Kelamin: Perempuan
Pekerjaan: Student
Hobi: Reading
Ras: Japanese
Status: Single
Agama: Buddha

Nama: Michiko
Umur: 20
Jenis Kelamin: Perempuan
Pekerjaan: Student
Hobi: Traveling
Ras: Korean
Status: Married
Agama: Buddha